

Lab 3: Give Solutions

Answer the following questions given in the book “ Introduction to the Theory of Computation by Michael Sipser”. (Pages 203-205)

Note: Although the solutions are given in the book, you are required to give the detailed solution by yourself. Copying the solution in the book will lead to zero marks.

- 4.5 Let X be the set $\{1, 2, 3, 4, 5\}$ and Y be the set $\{6, 7, 8, 9, 10\}$. We describe the functions $f: X \rightarrow Y$ and $g: X \rightarrow Y$ in the following tables. Answer each part and give a reason for each negative answer.

n	$f(n)$
1	6
2	7
3	6
4	7
5	6

n	$g(n)$
1	10
2	9
3	8
4	7
5	6

- | | |
|------------------------------------|------------------------------------|
| ^A a. Is f one-to-one? | ^A d. Is g one-to-one? |
| b. Is f onto? | e. Is g onto? |
| c. Is f a correspondence? | f. Is g a correspondence? |
- 4.6 Let $\mathcal{B}$ be the set of all infinite sequences over $\{0,1\}$. Show that $\mathcal{B}$ is uncountable, using a proof by diagonalization.
- 4.7 Let $T = \{(i, j, k) \mid i, j, k \in \mathcal{N}\}$. Show that T is countable.